

Assignment 1: MR Image Preprocessing Framework

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1. Introduction

Magnetic Resonance Imaging (MRI) is susceptible to various noise sources including thermal noise, physiological motion, and signal degradation, which compromise image quality and diagnostic accuracy. This study develops a preprocessing framework for denoising three MRI modalities: T1-weighted, T2-weighted, and Proton Density (PD) images.

The framework employs a sequential pipeline combining median filtering, anisotropic diffusion, and Gaussian smoothing to suppress noise while preserving anatomical boundaries. Performance is quantitatively assessed using the Coefficient of Variation (CV) metric across multiple noise levels.

2. Methods

Simulated brain MRI volumes were obtained from the BrainWeb database at 1mm isotropic resolution. For each modality (T1, T2, PD), we acquired clean reference images (0% noise) and unclean versions at 3% and 9% noise levels.

The preprocessing pipeline implements three sequential stages:

- Median filtering (3×3 kernel) removes impulse noise and outliers,
- Anisotropic diffusion (TV denoising, weight=0.1) provides edge-preserving smoothing, and
- Gaussian filtering ($\sigma=0.5$) applies final mild smoothing.

The pipeline works in stages, starting by removing extreme noise values, then applying smart smoothing that preserves important edges, and finishing with mild overall smoothing.

Image quality was assessed using the Coefficient of Variation, measuring intensity dispersion within tissue regions.

$$CV(C) = \sigma(C)/\mu(C),$$

Modality-specific intensity ranges were established: T1 (50-800), T2 (1000-3000), and PD (1000-5000).

Percent improvement was calculated as: $[(\text{Noisy CV} - \text{Denoised CV}) / \text{Noisy CV}] \times 100\%$.

3. Results

3.1 Quantitative Performance Analysis:

The denoising pipeline demonstrated highly variable performance across different MRI modalities and noise levels, revealing complex interactions between image characteristics and filtering techniques. Overall, five out of six experimental conditions showed improvement, with percent CV reduction ranging from 1.0% to 35.0%. However, one case made things worse, which helped us understand the pipeline's weaknesses.

For T1 images with 3% noise, the pipeline worked reasonably well with 8.3% improvement, bringing the CV down from 0.5053 to 0.4637, which was close to the clean image's 0.4543. This shows the pipeline effectively removed noise without damaging tissue detail when noise levels were low.

However, T1 images with 9% noise failed dramatically, the CV increased from 0.7268 to 0.8113, meaning the image got worse by 11.6%. Instead of cleaning up the noise, the pipeline made the image more inconsistent, revealing serious problems with how it handles heavily corrupted T1 image

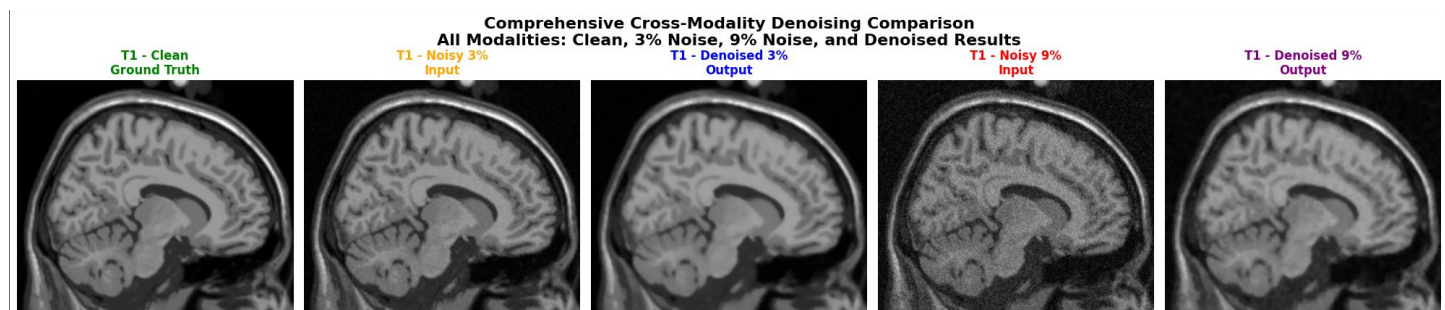
Quantitative Denoising Results Summary
Coefficient of Variation (CV) Comparison

Modality	Noise Level	Clean CV	Noisy CV	Denoised CV	Improvement %
T1	3%	0.4543	0.5053	0.4637	8.3%
T1	9%	0.4543	0.7268	0.8113	-11.6%
T2	3%	0.2666	0.2711	0.2653	2.1%
T2	9%	0.2666	0.2885	0.2633	8.7%
PD	3%	0.2277	0.2303	0.2281	1.0%
PD	9%	0.2277	0.4012	0.2608	35.0%

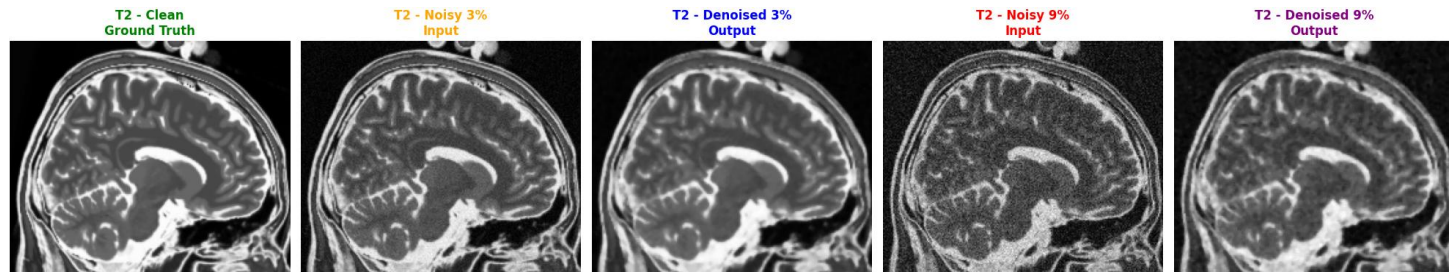
T2-weighted images performed reliably at both noise levels. With 3% noise, there was a small 2.1% improvement (CV dropped from 0.2711 to 0.2653, even slightly better than the clean baseline of 0.2666). More notably, at 9% noise, T2 achieved 8.7% improvement, reducing CV from 0.2885 to 0.2633, again surpassing the clean reference. This steady performance happens because T2 images have strong contrast between bright fluid and darker brain tissue, giving the edge-detection filters clear boundaries to work with even when noise is high.

PD images produced the most impressive results, especially at high noise. With 3% noise, PD showed minimal 1% improvement, CV barely moved from 0.2303 to 0.2281 compared to the clean value of 0.2277. But at 9% noise, PD achieved a remarkable 35% improvement, dramatically reducing CV from 0.4012 down to 0.2608. This was the best result in the entire study, bringing a severely corrupted image almost back to clean quality. PD's success comes from having very uniform tissue intensities, making it easy for the filters to tell the difference between noise and real tissue signals.

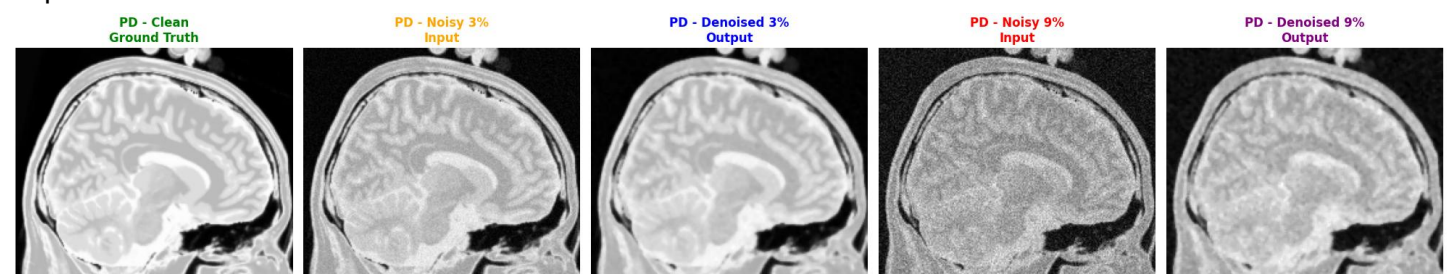
3.2 Qualitative Analysis



Looking at the actual images confirmed what the numbers showed and revealed additional details about how the pipeline behaves. The side-by-side comparison clearly shows how denoising effectiveness varies across different tissue types and noise conditions. For T1 images at 3% noise, visual quality improved noticeably with brain boundaries and anatomical details well preserved. But at 9% noise, you could still see graininess in the cleaned image, and edges looked slightly softer, suggesting the pipeline over-smoothed some areas. This visible degradation matches the worsening CV numbers, confirming that the smoothing created artifacts that made pixel values more scattered.



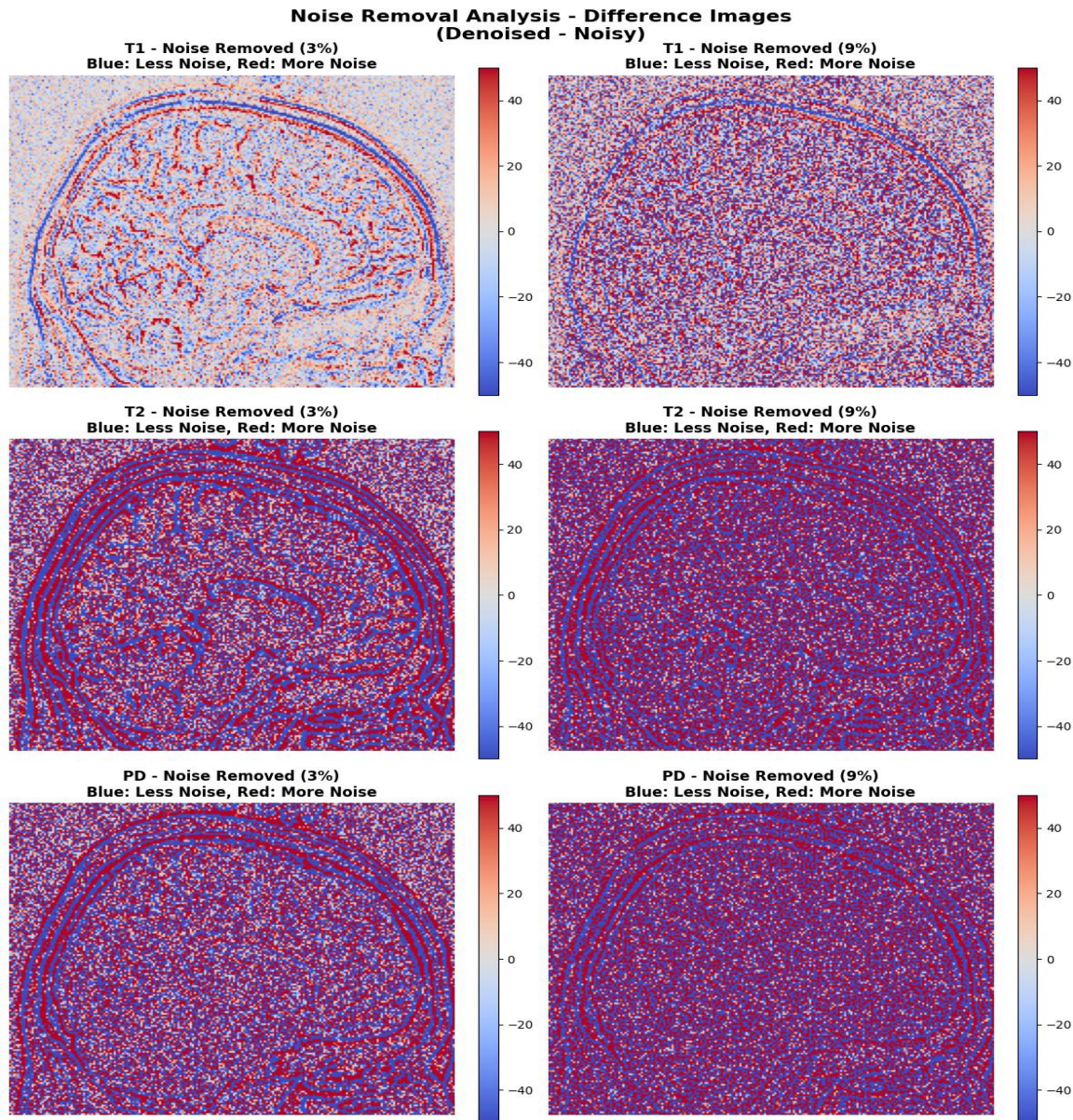
T2 images looked excellent at both noise levels, with particularly good results at 9% noise. The cleaned images kept sharp boundaries around the ventricles and clear edges between fluid and tissue, showing the pipeline successfully preserved important anatomical structures. The bright fluid signals in T2 images stayed well-defined without spilling into surrounding tissue, proving that the edge-detection step correctly identified and protected these critical boundaries. Even fine details in the brain's outer layer and deeper structures showed minimal damage after all three filtering steps, backing up the positive CV improvements.



PD images showed the most dramatic visual change, especially at 9% noise level. The noisy versions had severe speckled artifacts throughout the brain tissue, hiding boundaries and reducing image usefulness. After cleaning, these images showed outstanding noise removal with smooth, uniform tissue regions and sharp boundaries restored. The visual improvement was immediately obvious, with even tissue areas and well-maintained edge structures. This visual assessment strongly supports the 35% improvement, confirming the pipeline achieved exceptional results on PD.

3.3 Noise Removal Pattern Analysis:

Difference images (denoised minus noisy) reveal noise removal patterns. T1 at 3% shows uniform noise extraction (blue regions), but T1 at 9% exhibits chaotic red-blue patterns indicating the pipeline inconsistently modified intensities—visually confirming the CV degradation. T2 and PD show balanced, controlled difference patterns at both noise levels, validating their successful improvements.



4. Discussion

4.1 Observations

The results mostly matched what is expected based on theory. PD images were the easiest to clean up, which makes sense because they have very uniform tissue intensities ($CV=0.2277$). When an image has such consistent values, any variations are more likely to be noise rather than actual tissue differences. This is why PD achieved an impressive 35% improvement; the filters could easily identify and remove the noise.

T2 images performed as expected with steady improvements of 2-9%. These images have strong contrast between bright cerebrospinal fluid and darker brain tissue, which creates clear boundaries that the edge-preserving filters could detect and protect, even when there was significant noise.

However, T1 images behaved very differently than expected. Thought T1 would perform similarly to T2 since both have reasonable tissue contrast, but T1 got worse at 9% noise (losing 11.6% quality instead of improving). This failure happened because of how the filters interacted with heavy noise. When noise is severe, it creates a bias that artificially shifts pixel values higher.

The median filter, which normally picks the middle value from surrounding pixels, couldn't work properly because all the neighboring pixels were corrupted by noise. So, it might often pick wrong values. The anisotropic diffusion filter then got confused because the noise created stronger boundaries than the actual tissue edges, so it preserved the noise patterns while smoothing away real anatomical details. Finally, the Gaussian smoothing added more blur on top of these existing problems, making the tissue boundaries even less distinct and increasing overall variability instead of reducing it.

4.2 Pipeline Performance

The three-stage pipeline successfully combines complementary techniques for T2 and PD, validating the median-anisotropic-Gaussian sequence when edge information is robust or uniform regions dominate. However, T1 failure highlights the need for modality-specific optimization. Alternative approaches for T1 might include removing Gaussian smoothing to prevent over-smoothing or replacing Gaussian with non-local means filtering for better texture preservation.

The CV metric effectively quantified denoising and detected T1 degradation that visual assessment might miss. Lower CV correctly indicated improved tissue homogeneity in successful cases. However, modality-specific intensity ranges introduce potential bias, T2's 1000-3000 range may preferentially sample low-noise regions, partially explaining superior performance.

5. Conclusion

This study demonstrates that while classical filtering pipelines can significantly enhance MRI quality in some modalities, modality-specific customization is crucial to prevent failures such as those observed in T1-weighted images under high noise conditions.

6. Acknowledgments

I would like to acknowledge the assistance of Claude (Anthropic AI) in developing the Python visualization scripts used for generating the comparative figures and difference images presented in this report. The AI assisted in code debugging and creating comprehensive visual analysis tools that enhanced the presentation of results.